

Frontend Environment

Framework

- React.js (Web Dashboard)
- React Native (Mobile App)

Language

- JavaScript (or TypeScript for better scalability)

UI Framework

- Tailwind CSS
- React Native Paper / Material UI

State Management

- Redux Toolkit
- Context API

Routing

- React Router DOM

API Communication

- Axios

Form Validation

- React Hook Form
- Yup

Data Visualization

- Chart.js
- Plotly

Notifications

- Firebase Cloud Messaging (FCM)
- React Native Push Notifications

Mobile Alarm

- Android Alarm Manager
- React Native Background Timer

Development Tools

- VS Code
- Android Studio

- Expo CLI (optional)
- Node.js
- npm / Yarn

Frontend Folder Structure

frontend/

```
|
|
| └─ src/
|   |
|   | └─ components/
|   |
|   | └─ pages/
|   |
|   | └─ layouts/
|   |
|   | └─ services/
|   |
|   | └─ hooks/
|   |
|   | └─ utils/
|   |
|   | └─ assets/
|   |
|   | └─ redux/
|   |
|   | └─ context/
|   |
|   | └─ routes/
|   |
|   └─ App.js
|
|
| └─ public/
|
| └─ package.json
└─ README.md
```

Backend Environment

Language

Python 3.11+

Framework

FastAPI

API Documentation

Swagger UI

OpenAPI

Authentication

JWT

OAuth2

Password Hashing

bcrypt

ORM

SQLAlchemy

Database Migration

Alembic

Primary Database

PostgreSQL

Stores

- Users
 - Roles
 - Alarms
 - Reports
 - Scores
-

Secondary Database

MongoDB

Stores

- Challenge History
- Analytics Logs

- Recommendation Logs
- AI Predictions

AI Libraries

Scikit-learn

TensorFlow

PyTorch

XGBoost

Pandas

NumPy

Background Scheduler

Celery

Redis

(or APScheduler)

Notification Service

Firebase Cloud Messaging

Email Service

SMTP

SendGrid

File Export

ReportLab

OpenPyXL

Development Tools

VS Code

PyCharm

Postman

Docker Desktop

Git

GitHub

Backend Folder Structure

backend/

|

├─ app/

| └─ api/

| └─ auth/

| └─ models/

| └─ schemas/

| └─ services/

| └─ ai/

| └─ analytics/

| └─ recommendations/

| └─ database/

| └─ utils/

| └─ middleware/

| └─ config/

| └─ main.py

|

├─ requirements.txt

├─ Dockerfile

└─ alembic/

